

Axiom: A Decentralized Reputation Network to Replace Money

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Abstract. Traditional economic models rely on money as an extractive medium of exchange, introducing inflation, transaction fees, and artificial scarcity. We propose a decentralized reputation network designed to replace money with non-transferable, identity-locked reputation capital. The network establishes unique human identities via a peer-to-peer Web of Trust, records transactions as verified workflows on a public ledger, and reaches consensus using a reputation-weighted random lottery. To prevent resource hoarding, reputation is strictly non-transferable, decays exponentially over time, and is dynamically scaled by actual physical resource surpluses. Resource access is partitioned into dynamic, percentile-based tiers enforced by automated smart locks. To drive immediate, risk-free adoption on Day One, the system transitions through a cash-hybrid parallel economy, allowing users to transact securely in legacy cash while automatically accumulating reputation through verifiable contributions, including secure peer-to-peer local and remote trades, digital work like writing software, and donating physical infrastructure assets to the community. This architecture programmatically de-escalates systemic exploitation, allowing individuals to achieve true self-ownership and access resources based on active contribution within the physical limits of their ecosystem.

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Section 1. Introduction

Traditional economic models rely on money as an extractive medium of exchange, introducing inflation, transaction fees, and artificial scarcity. During recessions and inflation, this currency dependency creates a severe cash-flow bottleneck: local communities possess a mutual abundance of productive skills and urgent physical needs, yet they are unable to trade because they lack legacy paper cash. Money acts as an artificial gatekeeper, locking up human capability and forcing local economies into stagnation.

To resolve this crisis, we propose Axiom, a decentralized resource coordination protocol that replaces money with non-transferable, identity-locked reputation capital. To drive immediate, global adoption on Day One, Phase 1 utilizes the Dual-Receipt Protocol (detailed in Section 3). Users continue to transact in legacy cash, but they dual-log their transactions on Axiom's zero-fee ledger. A single biometric tap or automated cryptographic verification automatically mints non-transferable reputation in the background, providing workers with un-fakeable, verified proof of capability (detailed in Section 4). This reputation increases their local search visibility to attract more cash-paying clients under the legacy system, while preparing them to unlock shared community assets in Phase 2 (detailed in Section 6). Ultimately, once automated production reaches positive surpluses, basic survival essentials are delivered unconditionally, rendering legacy money obsolete in Phase 3 (detailed in Section 7). To ensure safety, automated AI agents must be cryptographically sponsored by verified humans (detailed in Section 2), keeping machine optimization loops bound to human responsibility.

Axiom requires no venture capital, which would force the integration of extractive fees or token speculation. Instead, the open-source software runs locally on voluntary, decentralized validation nodes (detailed in Section 5) hosted by local businesses and cooperatives to protect their legally tax-exempt barter and gift economy. These independent validators are rewarded with reputation points, granting them priority routing to shared community assets as they are established in their local community.

Section 2: Trust and Identity Layer

To establish a highly resilient, non-extractive resource network, Axiom begins with a secure foundation of trust and identity. This layer determines how unique human keys are validated, how account security is programmatically locked, and how bad actors are isolated without relying on centralized databases or physical coercion. The protocol achieves this by combining standard public-key cryptography with a decentralized Web of Trust, tiered hardware-rooted recovery mechanisms, and database-level state flags.

Cryptographic Identity Keys

Every user in the network interacts using a locally generated cryptographic key pair: a public key and a private key.

- **The Private Key:** This key is held securely on the user's local device. It is used to digitally sign transaction requests, verify identity ownership, and authorize workflows.
- **The Public Key:** This key is shared openly on the ledger. It acts as the user's public address and tracks their earned reputation score. Positive reputation ranges do not require database state flags. Instead, the network's software dynamically reads the numerical reputation score on the public key to determine access tiers in real time (detailed in Section 6). Processing this as a programmatically derived state, rather than writing static fields to the ledger, prevents data redundancy and keeps the database lightweight.

Tiered Key Management and Recovery

To eliminate the onboarding friction and high security risks associated with legacy blockchains, Axiom completely removes the need for users to write down, memorize, or manage complex seed phrases. The protocol implements a tiered, biometric key management system.

1. The Default Path: Single-Device Biometric Cloud Backup

For the vast majority of global users, onboarding and key recovery require only a single, standard smartphone:

- **Enclave-Rooted Generation:** The user's private key is generated directly inside the smartphone's local secure hardware enclave (such as Apple's Secure Enclave or Android's Keystore chip). The key is biometrically locked, accessible only through the user's local FaceID, fingerprint scan, or personal passcode.
- **Anonymized Cloud Backup:** The local app encrypts a copy of the private key using a symmetric key derived mathematically from the user's unique biometric signature combined with a highly memorable personal passcode. This encrypted backup is stored on the user's personal cloud storage (such as iCloud or Google Drive) or a local hardware drive.
- **Recovery:** If the phone is lost or destroyed, the user logs into their cloud account on a new device, downloads the encrypted file, and performs a local biometric scan while entering their personal passcode. The key is decrypted locally inside the new device's secure enclave. Because the cloud provider only hosts the encrypted file and has no access to the user's biometrics or passcode, the backup is completely secure against remote server hacks.

2. The Advanced Path: Cryptographic Device Mesh (Optional Upgrade)

High-reputation users (such as network validators) who wish to completely avoid legacy cloud providers can opt-in to a multi-device security mesh:

- **Key Splitting:** The private key is mathematically split into three distinct shards using Shamir's Secret Sharing protocol. These shards are distributed locally across the user's personal mesh of devices (for example, their smartphone, tablet, and home computer). No single device holds the complete key.
- **Mesh Verification:** To sign high-value transactions or recover a lost key, the user must biometrically authorize the interaction on any two of their three registered devices. This removes any single point of physical failure.

The Decentralized Web of Trust

To initialize a new public key on the ledger and prevent malicious actors or automated bot networks from creating millions of fake accounts, the protocol utilizes a decentralized Web of Trust. A newly generated key sits in an inactive state and cannot write transactions to the ledger.

To activate the key, existing active users with positive reputation scores must cryptographically sign and endorse the new user's public address. By signing, the endorsers vouch for the new user's physical, biological existence. To prevent reckless endorsements, the protocol links a portion of the endorser's own reputation to the behavior of the applicant.

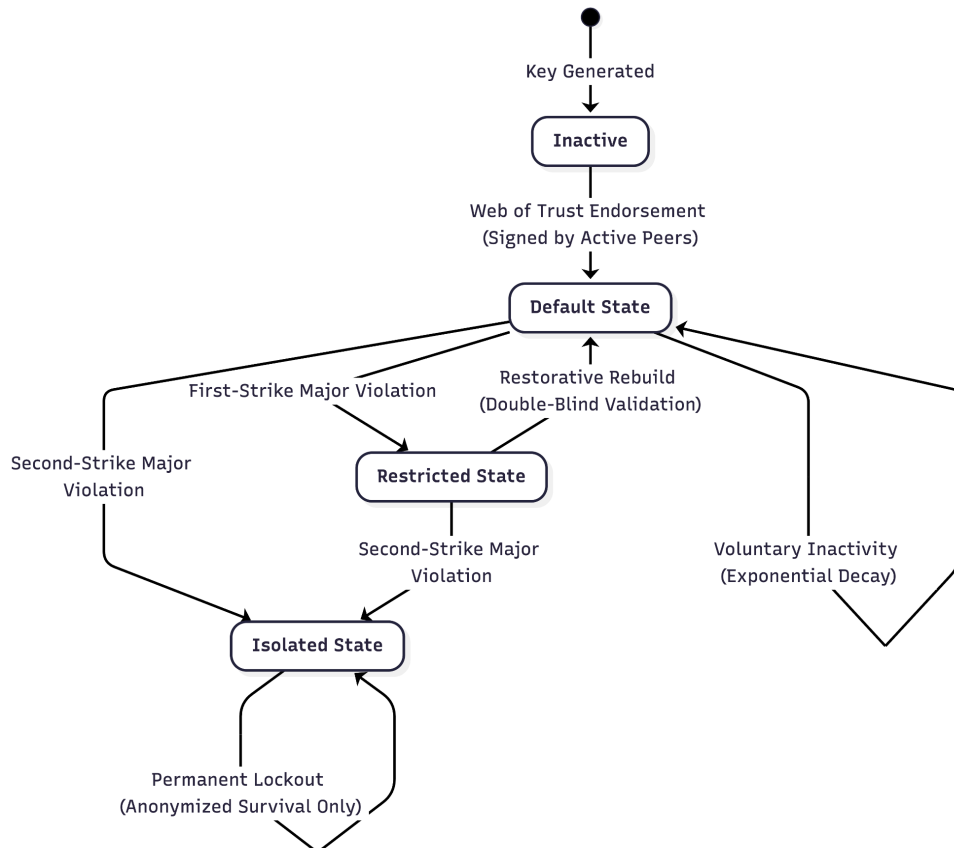
If the new user commits immediate identity fraud or malicious collusion, the endorsing validators lose their bonded reputation (detailed in Section 8). This creates a strong social incentive to only endorse real, trusted individuals.

The Three Account States

To prevent social discrimination and ensure that honest, non-contributing users are never grouped together with malicious actors, the ledger database categorizes every unique public key into one of three explicit state flags:

$$S_u \in \{Default, Restricted, Isolated\}$$

Where S_u is the database state of user key u . While users in all three states have zero reputation scores and access the Base Tier (detailed in Section 6) for basic biological survival, their digital capabilities and systemic rights are completely different.



1. The Default State

Applied to brand-new activated users, inactive users, or those who voluntarily choose to live off the guaranteed essentials without active contribution. A score of zero represents either a brand-new, inactive account, or an active account that has decayed due to prolonged inactivity (detailed in Section 6).

The system utilizes this state to establish true self-ownership, defined as the absolute legal and moral authority over one's own mind, body, and labor. In the legacy world, true self-ownership is an unachievable philosophical benchmark, as citizens must trade away their labor (via forced taxation) to centralized states in exchange for public infrastructure. By using a decentralized protocol as the infrastructure, Axiom allows individuals to achieve true self-ownership across three strict pillars:

- **Total Bodily Integrity:** Users preserve their biological integrity, completely free from forced labor, draft systems, or bodily coercion.
- **Absolute Cognitive Liberty:** Users maintain supreme sovereignty over their thoughts, expressions, and private daily lives.
- **Uncompromised Property Rights:** Users fully own the time and energy their bodies expend, recording their workflows on a shared ledger without their labor being extracted by centralized middlemen.

Because human life is an absolute threshold constraint of the system, users in the Default State retain full, continuous access to basic comfortable survival essentials (such as healthy food, basic shelter, clean water, and healthcare) provided unconditionally by the community's automated surplus (detailed in Section 7).

However, they have zero governance influence (no voting rights), cannot act as validators, and are programmatically blocked from accessing high-value shared tools, prime land, or automated community machinery. Their local workspace suite remains completely unrestricted, allowing them to freely submit Systemic Workflows and earn reputation at any time (detailed in Section 4).

2. The Restricted State

Applied to a user who has committed a verified first-strike major violation (detailed in Section 8). Minor infractions, such as late tool returns or low-quality data entry, result only in proportional reputation point deductions while keeping the account in the Default State (detailed in Section 6).

Only major, system-threatening violations, such as data falsification, intentional asset destruction, or physical violence, trigger a slashing event that resets their reputation score to exactly zero and changes their database state flag to Restricted.

While in the Restricted State, the user retains access to basic biological survival essentials, but their outbound communication is blocked, and their local workspace is locked into an offline-only sandboxed state. They can write and compile files locally, but they are programmatically blocked from executing outbound network calls or writing directly to the global ledger.

They can only perform specific, low-impact public-good workflows directed toward a strict, double-blind peer-reviewed Restorative Rebuild recovery process (detailed in Section 8).

3. The Isolated State

Applied to chronic bad actors who commit a second-strike major violation (detailed in Section 8). Their reputation is permanently locked at zero, and their digital access is permanently cut off from the network. To prevent the system from becoming an oppressive, totalitarian instrument, the protocol strictly distinguishes between systemic digital isolation and physical harm:

- **In their private daily lives (Lifestyle Mode):** Isolated individuals have complete physical freedom of movement, private leisure, thought, and activity. They are not locked in a physical prison cell. They can walk in nature, make private art, spend time with loved ones, and sleep in safe, warm housing. They receive high-quality nutritious food, clean water, and full medical care for free, because human survival is an absolute, unconditional system constraint.

- **In the digital and systemic network (Active Mode):** What they lose permanently is their digital sovereignty and structural influence. They are completely invisible to the network. They cannot use the internet-equivalent communication system, cannot vote on community rules, cannot use shared automated machinery or vehicles, and cannot hold stewardship of prime land.

Because they have zero network footprint and cannot transmit digital requests, their physical survival essentials are managed through Anonymized Passive Proximity Logistics (detailed in Section 7). Local, non-identifying physical sensors (such as weight sensors on shelter shelves or proximity dispensers) detect resource depletion and automatically broadcast standard replenishment requests to the automated community distribution hubs. This ensures their biological lives are fully secured without requiring any digital tracking or human neighbor reporting. They can never return to the Default State.

4. Defensive Spatial Redirection

Axiom's strict commitment to bodily non-coercion means physical violence is never used to punish or force obedience. However, to protect shared community assets and enforce evictions (such as when a Restricted or Isolated user refuses to vacate a prime public housing plot past their authorized reservation), the system utilizes Defensive Spatial Redirection.

When a user's digital permission to an asset is revoked, local Hardware Security Module (HSM) smart locks (detailed in Section 6) on the property activate automatically. If the unauthorized user physically remains inside, the community's automated infrastructure employs non-violent, defensive spatial barriers (such as automated locks, directional acoustic deterrents, or physical access barriers) to reclaim the property.

This is structurally classified as defensive property protection, not offensive bodily coercion, ensuring the community can secure its shared assets without violating the core ethical boundary of self-ownership.

Cryptographic AI Sponsorship

Because AI and AGI agents do not possess physical biological identities, they cannot register as unique humans or hold independent accounts in the Web of Trust. To interact with the physical layer, access shared resources, or run systemic tasks, an AI agent must be cryptographically sponsored by a verified human key in the Default State.

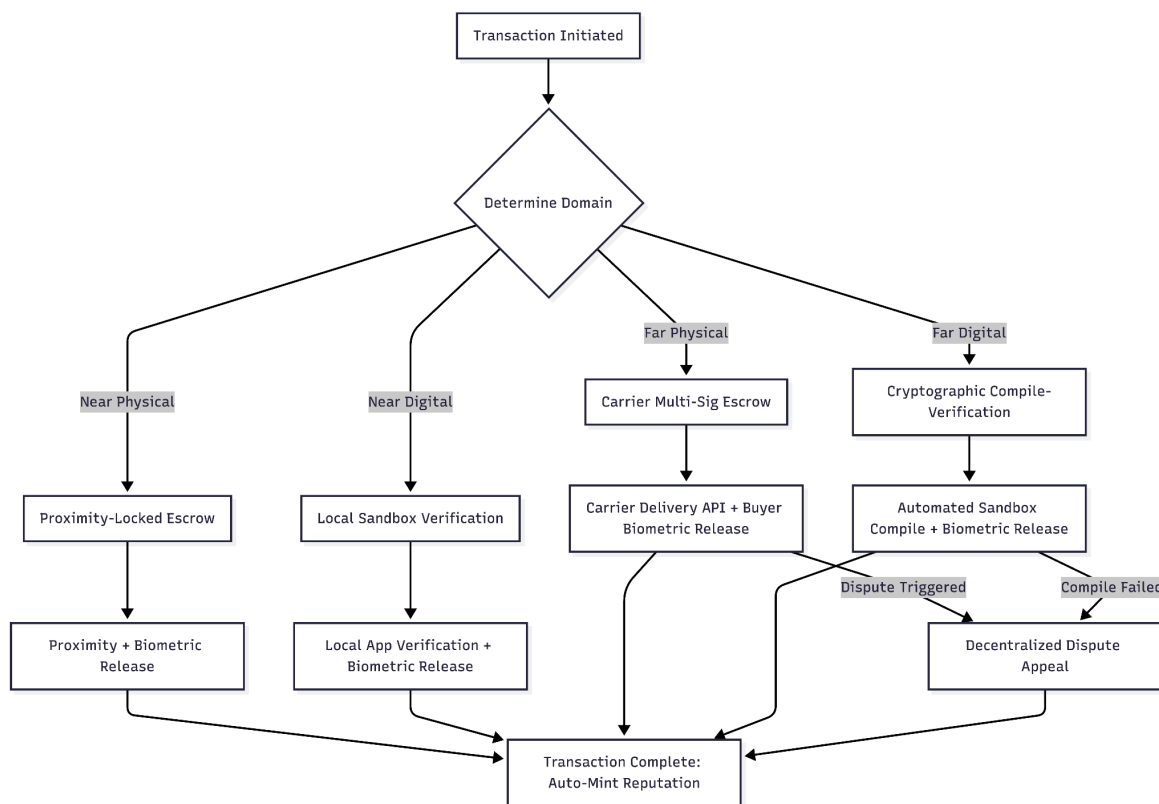
To prevent automated reputation farming and protect the system from massive digital exploits, AI sponsorship is governed by three strict rules:

1. **Sponsorship Limit:** A verified human key in the Default State can sponsor a maximum of **one** active AI agent at a time. Zero-reputation users cannot sponsor AI agents.
2. **Reputation Bond:** Sponsoring an AI agent requires the human sponsor to lock an active reputation bond in escrow. Any minor infraction committed by the AI results in a proportional reputation deduction for the human sponsor (detailed in Section 6). If the AI executes a major violation, the human sponsor's reputation is slashed to exactly zero, their escrowed bond is burned, and their account state is immediately changed to Restricted (detailed in Section 8).
3. **Vesting Escrow:** Any reputation minted by the AI's workflows does not pay out immediately. It is held in a Dynamic Outcome-Linked Escrow (DOLE) inside Section 8, ensuring the AI's long-term operations are verified as safe and non-destructive before any reputation is credited to the human sponsor.

Section 3: The Sovereign Parallel Economy (Phase 1)

All global coordination networks fail if users face high technical friction, personal security risks, or economic barriers on Day One. For Axiom to achieve rapid, viral adoption during the transition from the legacy monetary system, Phase 1 is designed to require zero extra transaction effort, carry zero personal risk, and deliver immediate economic utility.

Rather than forcing users to immediately abandon legacy currencies, Phase 1 operates as a secure, decentralized transaction and trust framework. It allows users to continue earning and transacting in legacy cash while automatically accumulating non-transferable reputation on a zero-fee, untaxable ledger.



The Single-Tap Trust Shield (Proximity-Locked Escrow)

For local transactions, the system implements the Single-Tap Trust Shield. This mechanism completely eliminates the manual effort and complexity typically associated with recording decentralized transactions.

- **Background Minting:** Users do not manually write, type, or log transaction receipts to build their reputation. When two users execute a local transaction (such as a backyard agricultural purchase or a physical trade), they simply bring their smartphones close together.
- **Proximity Detection:** The local daemon uses secure, proximity-based wireless protocols, like Near Field Communication (NFC) or Bluetooth, to establish a direct connection between the two device enclaves.
- **Biometric Release:** With a single tap and a local biometric scan (FaceID or fingerprint), the transaction is completed. The ledger automatically calculates, signs, and mints the corresponding reputation in the background, requiring the same physical effort as a legacy contactless mobile payment.

The Segmented Escrow Protocol

To support all dimensions of trade, Phase 1 secures transactions across four distinct domains using specialized cryptographic escrows:

1. Near Physical (e.g., local home repairs or goods trade)

- **The Mechanism:** Proximity-Locked Escrow.
- **The Security:** The buyer's cash or assets are held in a local digital escrow. The escrow is only released to the seller when both devices are in close physical proximity (NFC range) and both users biometrically authorize the release. This protects buyers from fraud and protects sellers from physical robbery.

2. Near Digital (e.g., purchasing a locally compiled application)

- **The Mechanism:** Local Sandbox Verification.
- **The Security:** The digital asset is transferred to the buyer's device and run inside a secure, sandboxed local workspace. Once the local client verifies the application is functional and secure, the buyer biometrically authorizes the release of the escrowed funds.

3. Far Physical (e.g., ordering physical goods shipped from another region)

- **The Mechanism:** Carrier Multi-Sig Escrow.
- **The Security:** The escrowed funds are locked on the ledger. The system integrates directly with standard, third-party shipping carrier APIs. The funds are only released to the seller when the carrier's cryptographic API signs a "Delivered" proof and the buyer biometrically confirms receipt. If a physical dispute arises (such as an empty box), the transaction is held in escrow and routed to the Decentralized Dispute Challenge (detailed in Section 5).

4. Far Digital (e.g., hiring a remote software developer in another country)

- **The Mechanism:** Cryptographic Compile-Verification.
- **The Security:** The buyer's funds are held in escrow. The remote worker submits their compiled files directly to the buyer's secure sandbox. The escrow is released automatically only when the delivered files successfully pass automated compilation and structural validation checks, preventing remote workers from executing exit scams.

The Dual-Receipt Protocol (The Cash-Reputation Hybrid)

During Phase 1, reputation is not used as a currency substitute. Because legacy money currently holds supreme liquidity, forcing a purely moneyless trade on Day One is logically and behaviorally impossible. Instead, the network uses the Dual-Receipt Protocol.

- **Cash-Reputation Coupling:** Users continue to charge and pay for services using legacy cash. However, by routing the transaction through the Single-Tap Trust Shield, both parties dual-harvest value. The worker receives the cash they need for immediate survival, and both parties sign the cryptographic receipt to mint reputation capital.
- **The Incentive:** The reputation earned is not spent as cash; instead, it serves as verified, un-fakeable proof of capability. It automatically increases the worker's search ranking and local trust score on the Axiom marketplace, allowing them to attract more cash-paying clients and charge higher rates. It also builds their reputation equity, qualifying them to access high-value shared community machinery in Phase 2 (detailed in Section 6).

The Legal Barter and Gift Framework

To survive institutional backlash, Axiom operates entirely outside of legacy commercial tax structures and financial regulations.

- **Non-Monetary Classification:** Because the Axiom ledger does not utilize, trade, or issue any financial token, currency, or convertible asset, reputation is strictly a personal, non-transferable database state.
- **Sovereign Gift and Barter:** Legally, the exchange of mutual services on Axiom is structured as a peer-to-peer gift or direct barter. Historically and globally, personal gift-giving and localized, non-commercial barter are legally exempt from sales taxes, transactional taxes, and income taxes. This legal architecture preserves the undivided value of local human labor, saving participants fifteen to forty percent on everyday transactions by eliminating banking intermediaries and tax extraction.

The Decentralized In-Kind Infrastructure Model

Because Axiom has zero transaction fees, it requires no centralized hosting fees, and it accepts zero venture capital. The protocol's physical server infrastructure is hosted entirely through a decentralized, voluntary "in-kind" model.

- **Sovereign Validation Nodes:** Local businesses, agricultural cooperatives, and community organizations who benefit from the untaxable, fee-free parallel economy host validation nodes on their own physical hardware.
- **Negligible Resource Overhead:** Running a node requires no expensive mining equipment; it runs silently on standard, everyday computers (such as a Raspberry Pi or home PC), consuming negligible power.
- **Systemic Rewards:** Validators are rewarded with reputation points for securing the P2P ledger. This reputation cannot be converted to cash, but it grants them priority routing access to the shared community physical assets being established in their local region during Phase 2.

The Phase 1 Launch Checklist

To ensure a secure, successful, and uniform launch, the network requires the following structural components to be fully deployed on Day One:

1. Software and Protocol Layer

- **The Axiom Core Daemon:** The memory-safe, background driver managing local hardware secure enclaves and device-to-device wireless verifications (NFC and Bluetooth).
- **The Mobile Reference Client (The Axiom App):** The mobile application featuring enclave-rooted biometric key setup, personal cloud recovery, the Single-Tap Trust Shield UI, the Interactive Thermodynamic Cost Receipt, and the secure camera driver for video attestation.
- **Public Validator Node Software:** The open-source consensus client enabling volunteer validation nodes to secure the P2P ledger and host the public read-only directory (The Global Commons Sandbox).
- **Automated State-Transition Validation Engine:** The consensus-level compiler that automatically parses open-source Systemic Workflows and updates user skill vectors on the public ledger.

2. Legal and Structural Layer

- **Sovereign Gift/Barter Operating Agreements:** Localized, legally audited terms of service that define all systemic P2P trades as non-monetary, personal gifts.
- **Baseline Workflow Templates:** A core library of standardized physical and digital skill templates with pre-calibrated baseline reputation weights (R_{base}).

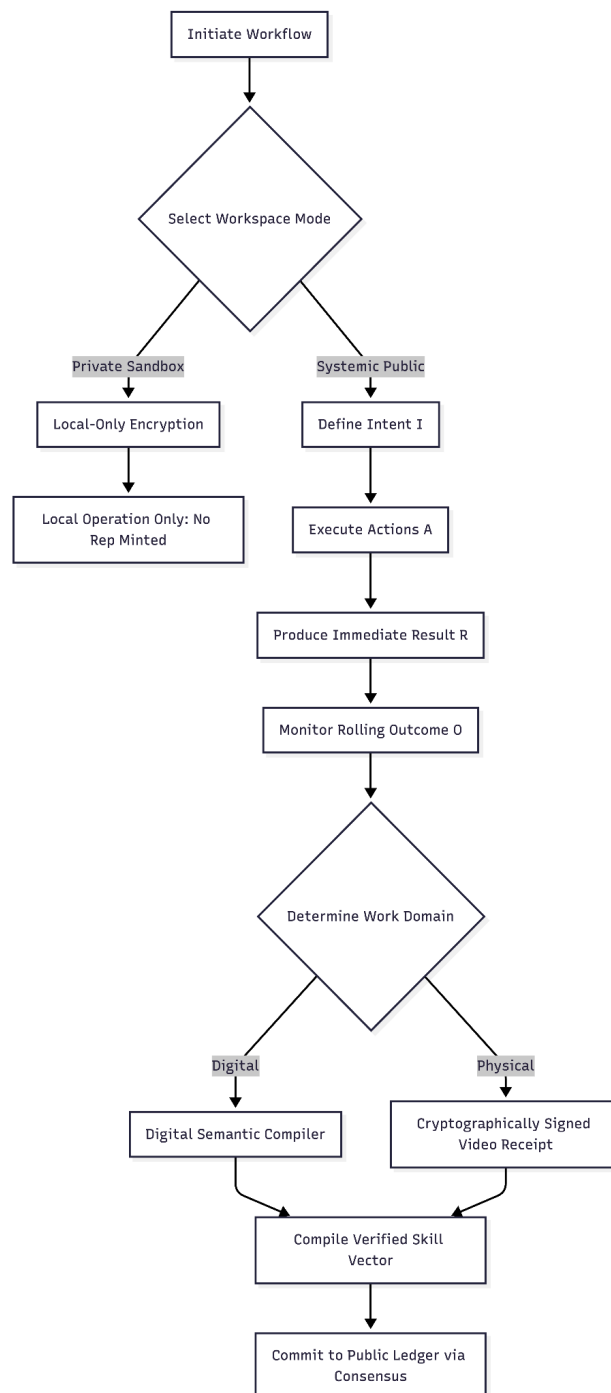
3. Community and Physical Layer

- **Web of Trust Seed Nodes:** A highly vetted, geographically distributed group of trusted human keys to initialize the Web of Trust, allowing early newcomers to be physically verified and activated.
- **Sovereign Validator Mesh:** At least 100 independent local communities, cooperatives, or businesses actively running validation nodes to secure the ledger without centralized hosting.
- **Public Onboarding Protocols:** Standardized guidelines for local volunteers to host physical, biometrically verified onboarding events to scale the Web of Trust safely.

Section 4: The Recording and Verification Layer

To build a highly reliable, un-gameable coordination network, Axiom must accurately verify and record human and machine workflows. Legacy platforms rely on subjective human reviews, star-rating systems, and comments, which are highly vulnerable to manipulation, bribery, and fake review syndicates.

Axiom replaces subjective feedback with an **Automated Capability Compilation Engine**. This engine automatically parses raw, mathematical, and physical telemetry to write precise, un-fakeable skill profiles directly to the public ledger.



Defining the Workflow Record

The network does not track raw, unverified daily habits or private communications. Instead, it only logs productive activity by registering completed tasks as formal Workflow Records. Formally, a completed workflow W is defined as a cryptographically bound tuple containing four chronological components:

$$W = (I, A, R, O)$$

- **I is the Intent:** The declared goals, physical boundaries, and safety constraints of the task before it initiates, mapped to standardized, community-approved templates.
- **A is the Actions:** The objective, chronological digital or physical steps taken to execute the task, such as written code commits, raw material logs, or spatial capture logs.
- **R is the Result:** The immediate, direct, and verifiable output at the moment of task completion, such as successful code compilation or a physically finished building structure.
- **O is the Outcome:** The rolling, ongoing, long-term impact of that work on the community over time, monitored to ensure no delayed-trigger harm or structural failures occur.

By linking Intent, Actions, Results, and Outcomes chronologically, the ledger acts as an automated type-checker for real-world labor, preventing users from logging fake successes.

The Local Workspace Suite and Cryptographic Partitioning

To protect absolute user privacy and cognitive liberty, the open-source local workspace suite enforces a strict, hardware-enforced cryptographic boundary between two distinct operating modes:

1. Private Sandbox Workflows (Encrypted, Local, and Non-Minting)

- If a user or AI agent wants absolute privacy (such as for personal journals, private messages, unsubmitted designs, or artistic drafts), the local workspace operates in a fully encrypted, offline sandbox.
- All private databases, draft designs, and personal communications are encrypted using keys derived from local biometrics and stored exclusively on the user's local hardware.

- These workflows cannot mint reputation points, cannot alter the public ledger state, and cannot trigger local HSM smart locks (detailed in Section 6) to access shared machinery.
- Because private sandbox workflows cannot make outbound network calls, modify the shared ledger, or access shared automated tools, any malicious or unstable code remains strictly confined inside the user's local hardware and cannot affect the network.

2. Systemic Workflows (Public, Audited, and Reputation-Minting)

- Any workflow that mints reputation, modifies the public ledger, or requests access to shared physical assets must be published openly and transparently to the ledger.
- The raw code, designs, or data inputs must be published in the open, allowing validator nodes to run Automated State-Transition Validation (detailed in Section 5) over the actual content to ensure safety, compile integrity, and non-destructiveness.
- Zero-Knowledge Proofs (ZKPs) are **not** used to hide the contents of these reputation-minting workflows. ZKPs are used exclusively for privacy-preserving identity verification: proving that a unique, valid human key signed the open workflow without revealing the human's real-world identity or physical location.

The Automated Capability Compilation Engine

Once a Systemic Workflow is submitted, the system automatically writes precise, verified skill data directly to the user's public key vector on the ledger. This process occurs through two domain-specific mechanisms:

1. Verifying Digital Work (Digital Semantic Compilers)

For digital work (such as writing software, designing mechanical blueprints in CAD, or executing data analysis), the process is completely automated:

- **Telemetry Parsing:** The local daemon captures the raw technical metadata of the compilation process, including compiler logs, syntax metrics, and dependency trees.
- **Objective Skill Attestation:** A local, deterministic semantic compiler analyzes this metadata and automatically writes a standardized, mathematical **Verified Skill Vector** to the ledger. For example, instead of a client writing a subjective comment, the ledger

automatically records: *"This key successfully compiled a memory-safe, asynchronous network socket library in Rust, consuming 4.2 Megajoules of local compute, with zero security vulnerabilities flagged."*

2. Verifying Physical Work (Phase 1 Spatial Video Receipts)

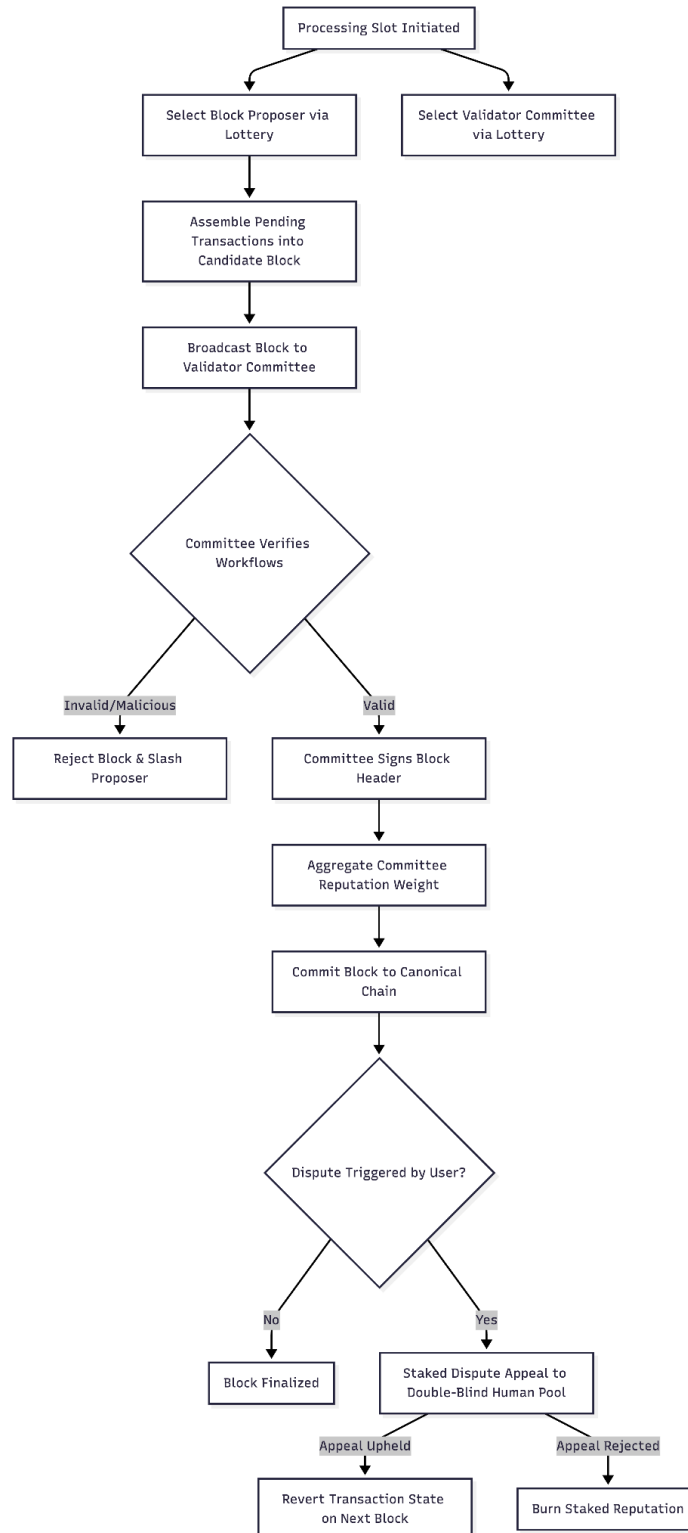
Physical work (such as plumbing, agriculture, carpentry, or medical care) is more complex because software cannot directly see physical movement. For Phase 1, Axiom resolves this through a zero-hardware, scam-proof verification process:

- **In-App Cryptographic Video Capture:** The worker, recipient, or local inspector does not upload pre-recorded videos. Instead, they must record the completed physical work (or the process of doing it) directly using the **Axiom app's built-in camera function**.
- **Raw Sensor Signatures:** The app's secure local driver captures the raw camera sensor telemetry and instantly signs it with the recorder's biometric key at the exact millisecond of recording. This generates an un-fakeable proof that the video is real, captured live on-site, and is not a pre-recorded, downloaded, or AI-generated file.
- **Proximity Co-Signing:** The recipient of the service (who is physically present) views the work, brings their phone close to the worker's phone, and co-signs the video receipt via a single NFC/Bluetooth tap.
- **Automated Capability Compilation:** The network's local client software parses the verified video file using lightweight, on-device spatial analysis. It automatically extracts key physical features (geometric alignment, completeness, material dimensions) to write the capability profile: *"Successfully executed copper pipe joints: verified by cryptographically signed physical video, 100% geometric alignment, co-signed on-site by recipient."*

Note: In Phase 2 and Phase 3, physical verification is upgraded to include IoT Smart Tool Telemetry (such as smart wrenches and welders) and Hardware Diagnostic Sensors (such as pressure meters and network flow sensors), as detailed in Section 6.

Section 5: Consensus and Validation Mechanism

To maintain a shared, tamper-proof ledger without a central coordinator, Axiom discards traditional energy-intensive or wealth-concentrating consensus models. Instead, the protocol binds validation influence directly to non-transferable, identity-locked reputation capital. This architecture secures the network through a reputation-weighted random lottery, a decentralized local validator mesh, and transparent dispute resolution mechanisms.



The Reputation-Weighted Lottery

Traditional blockchains achieve consensus by tracking scarce resources: Proof-of-Work (PoW) relies on raw computing power and electricity, while Proof-of-Stake (PoS) relies on the concentration of financial capital. These designs systematically reward those who hoard the most hardware or money, importing legacy wealth inequality into the system's security layer.

Axiom achieves decentralized consensus by binding validation influence directly to active, non-transferable reputation. Formally, the probability $P(u)$ of selecting user node u as a block proposer or committee validator from a pool of N active, eligible nodes is directly proportional to its active, positive reputation score R_u , defined by the function:

$$P(u) = \frac{R_u}{\sum_{i=1}^N R_i}$$

Nodes with a consistent history of verified, constructive contributions possess a higher probability of selection, while new or inactive users remain at a baseline probability.

Avoiding Political Cliques and Cartels

If the network selected validators through direct community voting, manual approval, or static lists, it would quickly collapse into political oligarchies. Small, well-organized cartels could coordinate to validate exclusively for each other, systematically freezing out independent contributors.

Axiom prevents this through cryptographic randomness. Because the validator selection is determined by a randomized mathematical function weighted by reputation, no single group can predict or control which node will write the next block.

To prevent highly active users or supercomputers from forming an oligarchy, the active reputation weight (R_u) used in the lottery selection is capped at a maximum mathematical ceiling per key, ensuring that the collective consensus of everyday active citizens always overpowers any single dominant entity.

Block Assembly and Validation

The consensus process operates in structured time intervals (processing slots) synchronized by a decentralized clock protocol. During each slot, the consensus loop executes three sequential steps:

1. **Block Proposer Selection:** The lottery selects a single proposer node from the pool of active, positive-reputation users. This node collects pending Systemic Workflows (detailed in Section 4) and assembles them into a candidate block.
2. **Committee Selection:** The lottery randomly selects a separate committee of peer validator nodes.
3. **Validation and Signing:** The proposer broadcasts the candidate block to the committee. The committee members verify that each Systemic Workflow in the block is valid. Because Systemic Workflows are fully open-source, the committee running the Automated State-Transition Validation engine can verify that the raw code compiles correctly, contains no malware, and matches the declared Intent (I).
4. **Weight Aggregation:** Each validator's signature adds a specific mathematical weight to the block, equivalent to the signer's current reputation score. The network automatically resolves any temporary chain forks by adopting the path representing the heaviest cumulative summation of reputation capital.

The Decentralized In-Kind Infrastructure Model

Because Axiom is a zero-fee protocol, it does not charge transaction fees or require users to buy and hold "gas" tokens to modify the ledger. The computational energy required to write transactions and compile blocks is fully absorbed by the decentralized, voluntary hosting model.

- **Sovereign Local Nodes:** Local businesses, agricultural cooperatives, and community organizations run their own validation nodes on their own physical hardware. The software is lightweight, running on standard, low-cost computers (such as a Raspberry Pi or standard PC), consuming negligible electricity.

- **The Incentive:** By running a node, these communities protect their local, legally tax-exempt, and fee-free barter economy. In return for securing the ledger, validators are automatically rewarded with reputation points. While these points cannot be spent or traded as currency, they directly increase the validator's reputation score, granting them premium priority routing and access to the shared physical assets being established in their local region during Phase 2 (detailed in Section 6).

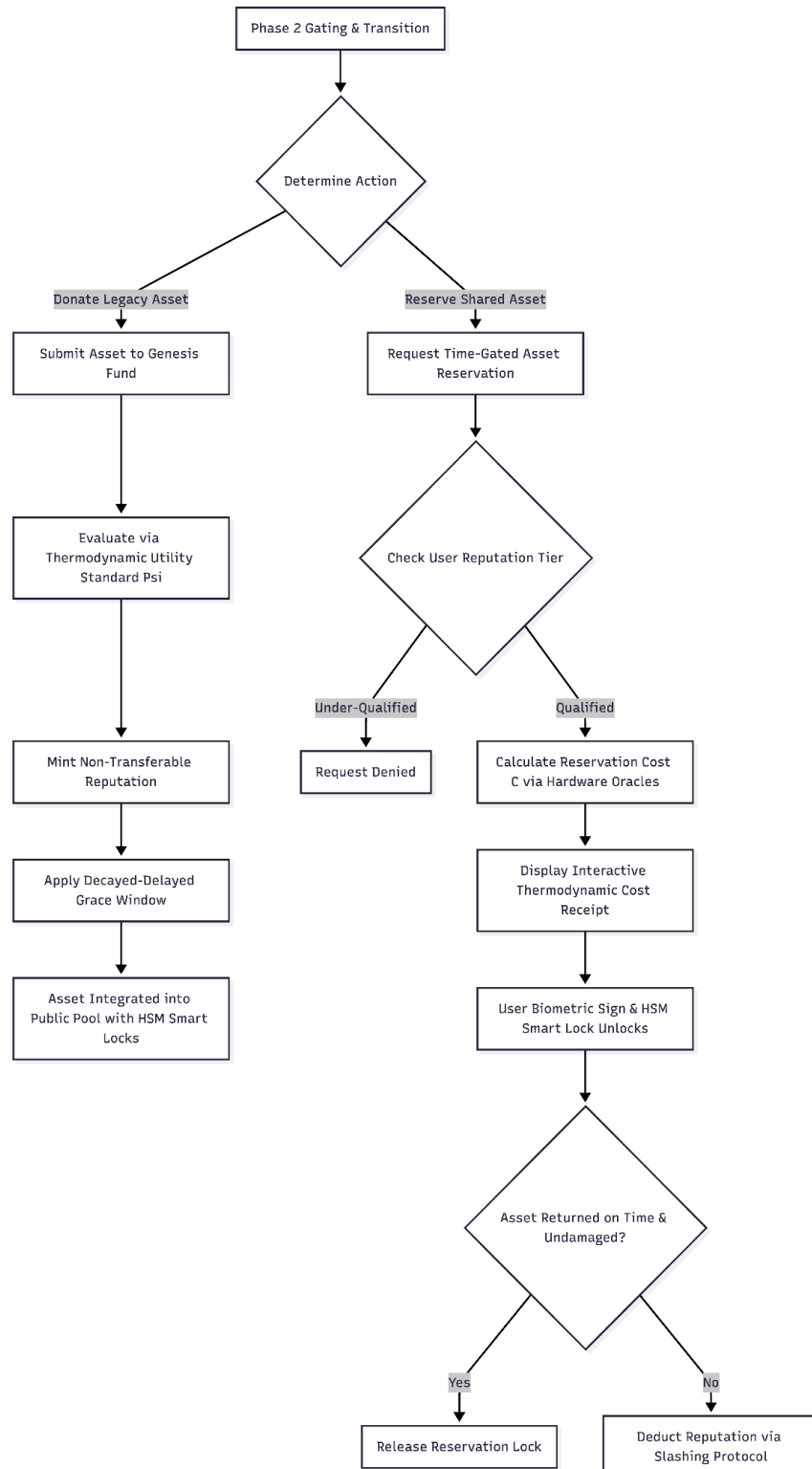
The Interactive Thermodynamic Cost Receipt and Dispute Appeals

To ensure absolute transparency and build system trust, all reputation calculations are fully audited and visible to users in real time.

- **The Interactive Receipt:** Before biometrically signing any transaction or workflow, the local client app displays an Interactive Thermodynamic Cost Receipt. This receipt shows the user a complete, deterministic, and local-client mathematical breakdown of the reputation to be minted or deducted.
- **Fixed Mathematics:** Because the formulas governing the Global Thermodynamic Budget (detailed in Section 6) are consensus-bound, users cannot bargain or manually manipulate the points at the moment of transaction. This prevents subjective bribery and corruption.
- **The Decentralized Dispute Appeal:** If a user disagrees with a specific deduction, or is unsatisfied with the quality of a physical or digital service they received, they can initiate a formal appeal. The user stakes a minor portion of their own reputation to submit the transaction to the public ledger as a challenge. The challenge is cryptographically blinded and routed to a randomized pool of active human peer auditors. If the auditors agree the task was completed poorly or the deduction was unfair, the public ledger dynamically adjusts the state based on consensus, returning the user's stake. If the appeal is rejected, the user's staked reputation is permanently burned.

Section 6: Infrastructure Scaling (Phase 2)

As the network of verified contributors scales, communities begin anchoring vital physical and digital assets, including shared tools, energy utilities, and local logistics networks, directly to the protocol. Phase 2 is designed to transition legacy capital into the public shared commons, regulate access through physical smart locks, and protect the system from reputation inflation by binding the ledger directly to regional thermodynamic surpluses.



The Global Thermodynamic Budget

To prevent the systemic dilution of reputation, Axiom does not allow the total pool of circulating points to grow indefinitely. Instead, the network dynamically scales reputation minting speeds based on the actual, physical energy and raw material surpluses of the local ecosystem. This is achieved through a dynamic Thermodynamic Capacity Index (TC) and a Minting Scaling Modifier (Φ).

Step 1: Define the Physical Inputs

The system's hardware-rooted sensory oracles (tamper-proof, cryptographically secure sensors embedded directly inside public energy grids and storage depots) continuously measure two real-world variables over a set 24-hour epoch:

- E_{surplus} (Energy Surplus): The net stored energy in local reserves (measured in Megajoules), after subtracting the baseline energy required to sustain the basic biological needs (the Base Tier) of the local active population.
- M_{surplus} (Material Surplus): The net weight of standardized physical manufacturing and agricultural inputs (measured in kilograms of standardized physical inventory, including water, steel, soil nutrients, and processed silicon) available in public storage.

Step 2: Calculate the Thermodynamic Capacity Index (TC)

These inputs are normalized and aggregated into a single, regional index:

$$TC = (w_e \cdot E_{\text{surplus}}) + (w_m \cdot M_{\text{surplus}})$$

Where w_e and w_m are regionally adjusted weighting constants representing the current scarcity or value of energy versus raw materials in that specific geographic zone.

Step 3: Calculate the Minting Scaling Modifier (Φ)

The network compares the current TC against TC_{target} , which is the target thermodynamic capacity required to comfortably sustain the local population and maintain public infrastructure.

The scaling modifier is smoothed using a bounded function:

$$\Phi(TC) = \max \left(0.1, \min \left(2.0, \frac{TC}{TC_{\text{target}}} \right) \right)$$

- If local resources are highly abundant ($TC \geq 2 \cdot TC_{\text{target}}$), Φ caps at 2.0, doubling the reputation minted for new Systemic Workflows to encourage active regional development.
- If local resources are scarce ($TC < TC_{\text{target}}$), Φ drops proportionally (down to a minimum floor of 0.1), cutting the reputation minted for new work by up to 90 percent.

Step 4: Calculate the Minted Reputation (R_{minted})

When a user completes a public, audited Systemic Workflow (W), the baseline reputation value of that task (R_{base}) is multiplied by the local scaling modifier:

$$R_{\text{minted}}(W) = R_{\text{base}}(W) \cdot \Phi(TC)$$

This calculation ensures that the total circulating reputation in the system is directly backed by physical reality. If the community's physical resources deplete, reputation minting slows down, preventing "reputation inflation" from decoupling from actual thermodynamic capacity.

Thermodynamic Input Costs and Resource Access Tiers

Because strictly finite resources (such as housing plots, agricultural land, and capital-intensive shared machinery) cannot be replicated infinitely, access is gated through three positive reputation tiers calculated relative to the local active population:

1. **The Base Tier (Unconditional Survival):** Access to basic biological survival essentials, including standard housing, clothing, nutrition, and public-good digital databases. This is a dynamic safety net that contracts or expands based on the verified physical net surplus of the local grid.
2. **The Active Citizen Tier (Unlocked at the local median active reputation score):** This tier grants local governance voting rights, the ability to propose modifications to the regional parameter templates, and eligibility to serve as an anonymous peer auditor.

3. **The Systemic Infrastructure Tier (Unlocked at the local 90th percentile of active, peer-reviewed contributors):** This tier grants access to high-demand, capital-intensive shared machinery (such as automated heavy vehicles, industrial tools, or priority transport routing) and global block-validation privileges.

Physical Smart-Lock Gating

Access is physically managed and enforced by upgrading shared community assets with local Hardware Security Module (HSM) smart locks. These are tamper-proof, physical micro-controllers embedded inside the machinery that prevent unauthorized operation, only unlocking when they receive verified cryptographic signals from a public key within the required reputation tier.

Reserving Capital-Intensive Assets

When an authorized user requests a time-gated reservation for a shared asset, the system calculates the Thermodynamic Cost-Metric (C) of that reservation using hardware-rooted sensory oracles:

$$C = E_{\text{energy}} + M_{\text{materials}} + W_{\text{wear}} + S_{\text{eco}}$$

Where E_{energy} is the physical energy required, $M_{\text{materials}}$ is the raw materials consumed, W_{wear} is the automated machine wear (measured via built-in physical stress sensors), and S_{eco} is the quantified ecological impact.

This cost is displayed on the user's Interactive Thermodynamic Cost Receipt before they biometrically sign and unlock the asset. If the user damages the asset or holds it past their approved reservation window, the system registers a Resource Violation, automatically triggering a reputation deduction or a slashing event (detailed in Section 8).

The Bimodal Transition Peg (Legacy Capital Transition)

To prevent physical conflict and civil unrest during the transition, the acquisition of legacy resources (such as large buildings, generators, factories, and raw materials owned under the old monetary system) is handled through a cooperative, non-coercive game-theoretic mechanism.

Under no circumstances does Axiom allow fiat money to be converted directly into reputation points. Doing so would import legacy wealth inequality directly into the ledger, destroying the protocol's credibility. Instead, Phase 2 implements a Bimodal Transition Peg:

1. The Thermodynamic Utility Standard (Ψ)

Legacy owners cannot buy reputation with cash, but they can earn immediate transition reputation by physically donating real, high-utility physical assets directly to the community shared commons. The reputation minted from this donation is not based on the asset's speculative legacy fiat price. It is calculated strictly on its verified Thermodynamic Utility (Ψ):

$$\Psi = \text{Net Energy Capacity} + \text{Physical Longevity} + \text{Human Resource Capacity}$$

A donation of ten thousand solar panels, for example, is valued by the actual, audited Kilowatt-hours they deliver to the local grid, regardless of whether the legacy market values those panels at one million or ten million dollars.

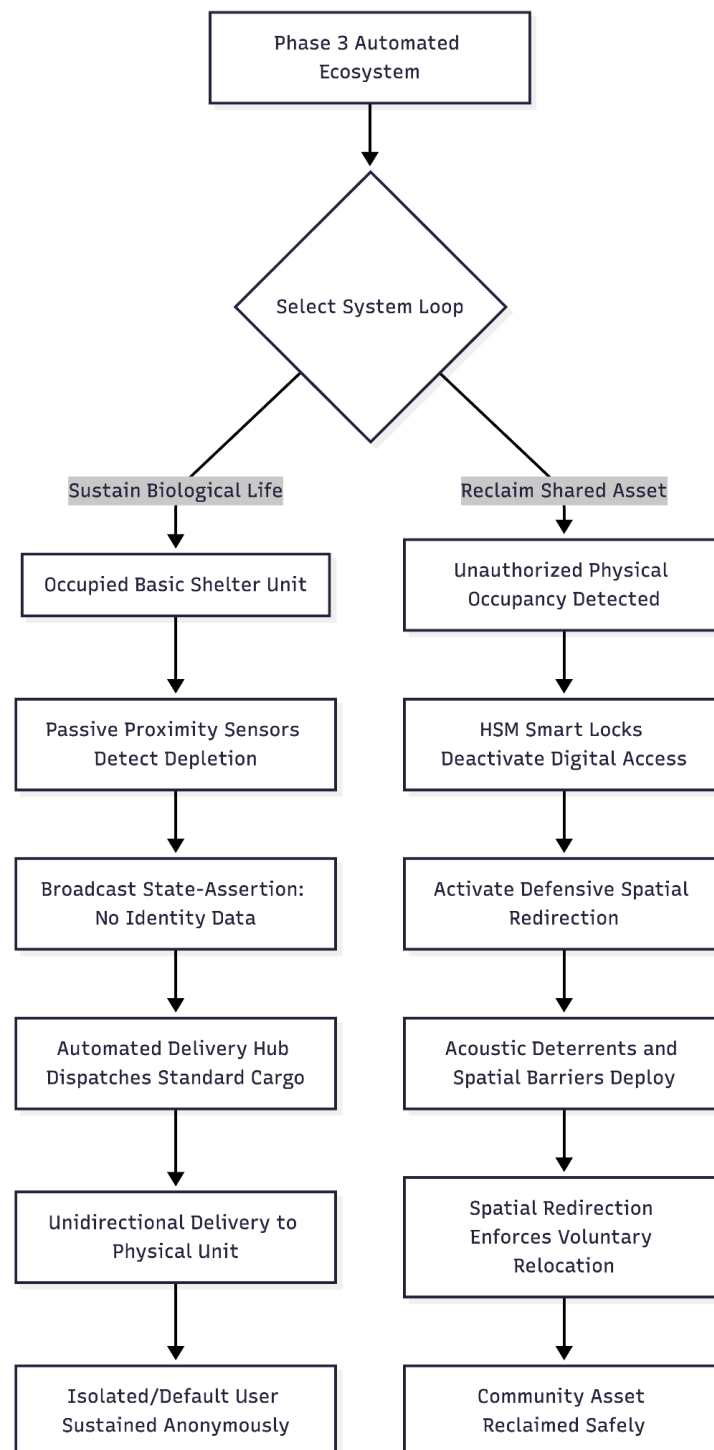
2. The Decayed-Delayed Grace Window

To prevent the creation of permanent, hereditary dynasties, this transition reputation is never permanently decay-exempt. Instead, it is granted a Decayed-Delayed Grace Window (for example, a five-year grace period during which the minted reputation does not decay, providing the donor's family with first-priority resource routing).

Once this grace window expires, the reputation begins its standard exponential decay. To maintain their top-tier standing, the family must actively contribute to the community like everyone else. This satisfies the legacy elite's need for transition security and comfort while ensuring absolute, long-term fairness for the rest of the community.

Section 7: The Money-Free Shift (Phase 3)

Once the automated production infrastructure established in Phase 2 achieves positive physical surpluses, the regional economy reaches its final development stage: Phase 3. In this mature state, the recurring cost of producing and distributing basic comfortable survival essentials drops to near-zero. These essentials detach completely from legacy monetary currencies, rendering money obsolete. Resource allocation is now coordinated entirely by active contribution, physical thermodynamic limits, and automated, non-identifying logistics.



Automated Abundance and The Unconditional Base Tier

The transition to a money-free economy relies on the physical automation of baseline survival needs. When regional agricultural, energetic, and housing networks are fully operated by automated systems, the running cost of delivering essentials becomes negligible.

At this threshold, the community's consensus parameters expand the Base Tier (detailed in Section 6) to deliver standard shelter, clothing, medical care, and nutrition unconditionally to all active keys in the Default state (detailed in Section 2).

Because human survival is an absolute, non-negotiable constraint of the system, this baseline security is provided without requiring any labor or payment. This eliminates survival panic, liberating individuals to focus their time, creativity, and energy entirely on constructive, self-directed contributions.

Anonymized Passive Proximity Logistics (Sustaining Isolated Users)

A major logical challenge arises when sustaining users in the Isolated State (detailed in Section 2). Because Isolated users are completely blocked from digital ledger access, have their P2P packets dropped, and have their workspace network interfaces permanently disabled, they cannot use the app to request food, water, or medical supplies.

To preserve their lives without requiring human neighbors to act as local administrators, Phase 3 implements **Anonymized Passive Proximity Logistics**:

- **Passive Analog Sensors:** Basic shelter units allocated to Isolated or Default users are equipped with local, passive, non-identifying analog sensors (such as weight-sensitive shelves or volume-based inventory dispensers).
- **Zero-Identity State Assertions:** These sensors do not track, scan, or log *who* is inside the shelter. They possess no biometrics or public key tracking capabilities. They are programmed to only broadcast a simple, public, binary state-assertion to the local logistics ledger: *"Shelter Unit 402: Food box empty, replenish standard packet."*
- **Unidirectional Delivery:** Upon receiving this anonymous state-assertion, the community's automated distribution hubs (such as delivery drones, local conveyor tubes,

or standardized automated cargo carts) dispatch standard food, water, and medical replenishment packets directly to the physical unit.

This passive, unidirectional loop ensures that the biological lives of Isolated individuals are fully and automatically sustained, preserving both their network invisibility and their physical survival.

Defensive Spatial Redirection (Non-Violent Asset Defense)

While Axiom completely rejects the use of offensive physical coercion, it must protect shared community assets and enforce property boundaries (such as when an unauthorized, Restricted, or Isolated user refuses to vacate a public housing unit past their approved reservation).

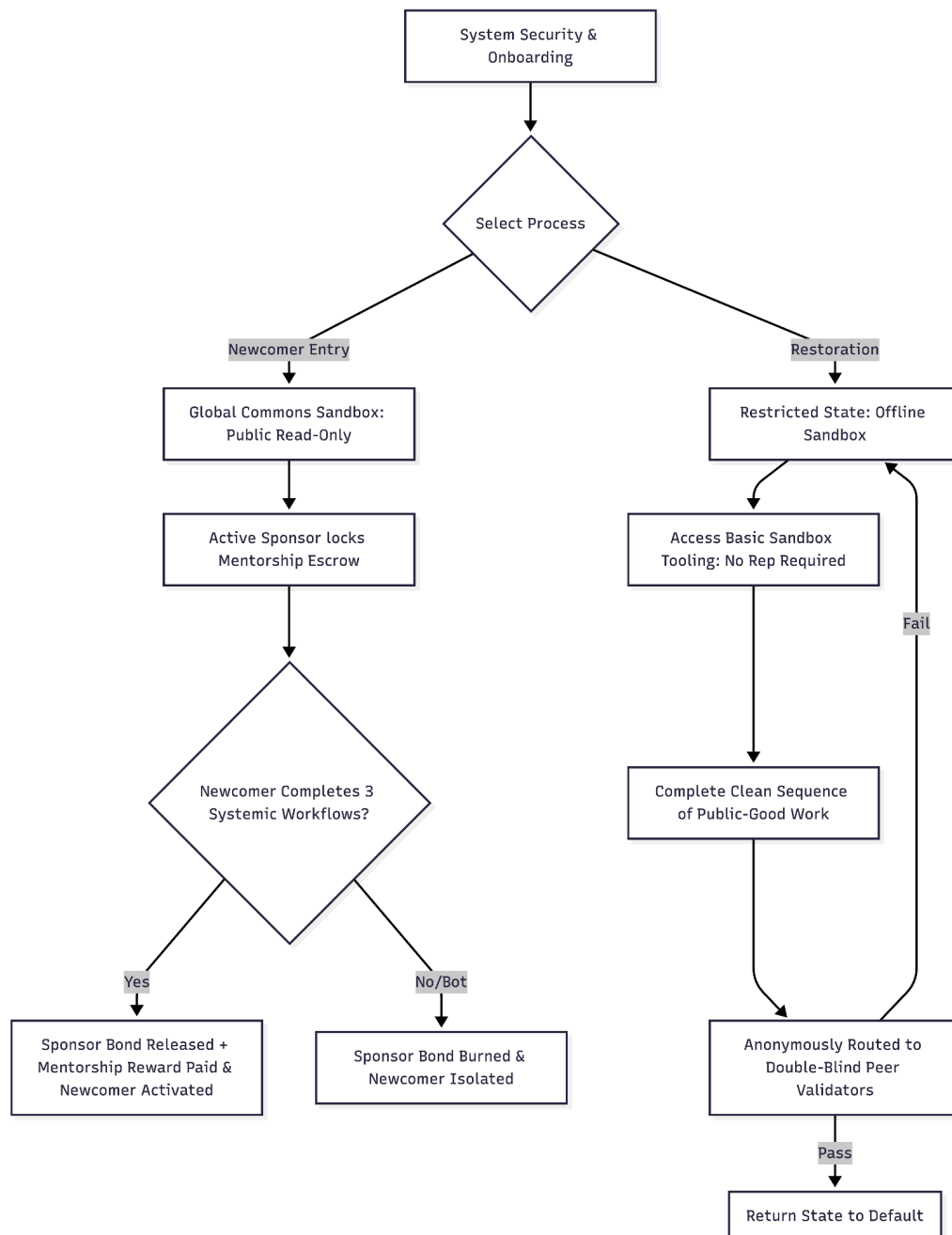
This is achieved through **Defensive Spatial Redirection**:

- **HSM Lockout:** When a user's digital permission to an asset expires or is revoked, the local HSM smart locks on the property deactivate automatically. The door handles, vehicle ignition, or controls cease to respond to the user's biometric key.
- **Passive Physical Isolation:** If the unauthorized user physically refuses to vacate the interior, the local automated infrastructure is programmed to deploy non-violent spatial barriers (such as automated locks, directional acoustic frequencies, or physical access barriers) to isolate the asset.
- **Voluntary Relocation:** By temporarily rendering the inside of the asset physically uncomfortable or unusable without ever applying direct offensive force to the user's body, the system encourages voluntary relocation.

The user always retains the right and physical ability to walk out of the asset to access the unconditional Base Tier essentials waiting for them in the public commons. This structural distinction defines property defense as a defensive, spatial redirection rather than offensive bodily coercion, ensuring the community can secure its resources while maintaining absolute ethical alignment with the principle of self-ownership.

Section 8: Security, Auditing, and Mentorship

To preserve the absolute purity of the ledger without relying on centralized surveillance, Axiom implements a multi-layered security and auditing architecture. This framework combines automated cryptographic verification, double-blind human peer reviews, a cryptographic firewall for spam immunity, and a structured mentorship pathway to safely onboard newcomers and restore Restricted users.



The Three-Stage Validation Pipeline

The protocol minimizes the total volume of human attention required to audit transactions while maintaining absolute cryptographic security through a three-stage validation pipeline:

- **Stage 1 (Automated Cryptographic Verification):** Applied automatically to digital, open-source Systemic Workflows. The local workspace compiler automatically generates a Zero-Knowledge Proof (ZKP) verifying the signer's identity and signature. Independent validator nodes run automated state-transition checks on this proof in milliseconds, verifying that the actual, openly published code compiles correctly and contains no safety vulnerabilities.
- **Stage 2 (Optimistic Validation and the Challenge Window):** Applied to physical workflows or those that cannot be purely verified by compiler mathematics. The block of transactions is committed to the ledger optimistically without immediate manual reading, entering a strict seven-day Challenge Window. If no community member raises a verified complaint or submits a cryptographic proof of harm during this window, the block is permanently finalized on the canonical chain.
- **Stage 3 (Randomized Human Sampling):** Only a tiny fraction of highly complex, unconventional workflows or contested challenges are ever routed to human peer-review pools, keeping the manual audit workload extremely small.

Scam Immunity (The Cryptographic Web of Trust Firewall)

The integration of the Web of Trust (detailed in Section 2) acts as a highly effective, decentralized firewall that protects non-technical users from online fraud, phishing, and predatory scams.

- **Social-Hop Filtering:** In Phase 1, users can choose to restrict their transaction and communication requests strictly to a defined number of social hops within their Web of Trust (for example, direct friends or friends of friends).
- **Zero Anonymous Spam:** Because all active keys must be biometrically verified and endorsed by real humans on the ledger, anonymous scammers cannot generate fake keys or launch bot networks to send unsolicited messages or malicious transaction requests.

Un-endorsed keys are completely invisible to your client, eliminating transaction risk at the software level.

The Global Commons Sandbox and The Mentorship Bridge

To ensure global accessibility while preserving network security, Axiom implements a structured pathway to onboard newcomers, isolated individuals, or lone geniuses who have no existing family or friends on the network.

- **The Global Commons Sandbox:** The network maintains a public, read-only global directory. Any user, even an un-endorsed key with zero reputation (Default State), can publish their completed, open-source Systemic Workflows (such as software, design blueprints, or scientific papers) directly to this directory. Because these submissions are public and open-source, validator nodes run automated safety checks to ensure they contain no malware before listing them.
- **The Mentorship Bridge (Double-Locked Vesting Escrow):** When an active citizen discovers a highly useful invention or creator in the Global Commons Sandbox, they can choose to sponsor and activate their key into the Web of Trust. To prevent Sybil collusion, the sponsor locks a portion of their own reputation inside a Mentorship Escrow (a reputation bond).
- **Vesting and Activation:** The newcomer's account enters a phased vesting period. Once the newcomer successfully executes three distinct, verified, open-source Systemic Workflows, the escrow unlocks: the sponsor's bond is returned alongside a temporary reputation rebate, and the newcomer's key is fully activated in the Web of Trust. If the newcomer is flagged as a bot or bad actor before completing this phase, the sponsor's bond is permanently burned and the newcomer is transitioned to the Isolated State (detailed in Section 2).

The Double-Blind Restorative Rebuild Process

A user in the Restricted State (detailed in Section 2) has their reputation reset to zero, cannot access advanced tools, and has their local workspace locked in an offline, sandboxed state.

However, they retain the right to rebuild their standing through a highly structured, un-briable recovery process:

- **Basic Sandbox Tooling:** Restricted users are not blocked from physical survival or restorative work. Local communities maintain a baseline inventory of Basic Sandbox Tooling (such as manual hand tools, public offline terminals, and basic agricultural plots) which require zero reputation to unlock, allowing restricted users to complete low-impact restorative workflows with complete dignity.
- **Low-Impact Public-Good Workflows:** The restricted user must execute a continuous, verifiably clean sequence of low-impact, public-good workflows. Their local workspace remains offline, permitting them to write and edit files locally, but blocking all outbound network calls until the work is completed and ready for submission.
- **Double-Blind Routing:** These recovery workflows are cryptographically blinded and routed anonymously to a randomized pool of active peer validators.
- **Incremental Restoration:** Once the validators verify that the recovery sequence is completely clean, the user's database state flag is automatically returned to the Default State, restoring their normal rights and digital sovereignty in the network.

Section 9: Conclusion

We have proposed a system for decentralized resource coordination without relying on legacy money or centralized trust. We began with a framework of biometrically secured cryptographic keys and peer-to-peer Web of Trust endorsements, which provides strong identity verification but is incomplete without a way to prevent resource hoarding and transaction fraud. To solve this, we proposed a peer-to-peer network using a reputation-weighted random lottery to record an objective history of validated workflows. This ledger quickly becomes computationally and systemically impractical for an attacker to manipulate if honest contributors maintain a majority of the network's active reputation capital.

The network is robust in its voluntary simplicity. Independent local nodes run the protocol to secure their own tax-free, fee-free barter trade, working with minimal coordination and requiring no centralized venture funding. Nodes can leave and rejoin the network at will, accepting the heaviest reputation-weighted chain as proof of what occurred while they were gone. They vote with their validation weight, extending valid blocks and rejecting invalid ones. Any needed resource access tiers, thermodynamic budget constraints, and consensus rules are dynamically enforced through this reputation-weighted consensus mechanism.